INCH-POUND
MS27151C
15 April 2022
SUPERSEDING
MS27151B
19 January 1976

DETAIL SPECIFICATION SHEET

NUT, STAMPED – STANDARD TYPE

Reactivated after 15 April 2022 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the described herein shall consist of this specification sheet.

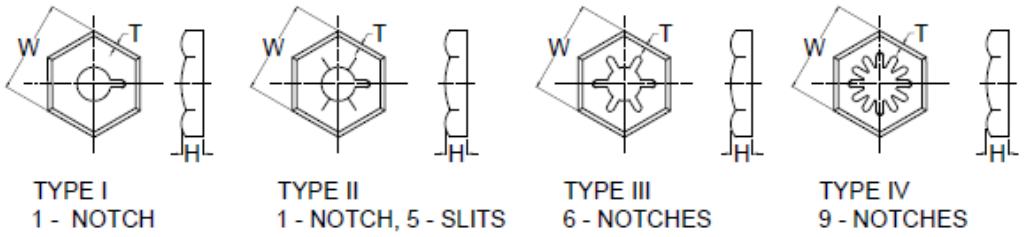


FIGURE 1. Stamped nut – standard type.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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TABLE I. Dash numbers and dimensions.

Dash No.	Nominal Size And Threads	Series	Type	W Across Flats		H Height		T Thickness ± .002	Install Torque-Ref (In.-Lbs)	Tension Ref (Lbs)
				Max	Min	Nom	Tol			
1*	(3) 0.099-48UNC	Mach. Screw	I	0.187	0.180	0.068		0.009	2	60
2	(4) 0.112-40UNC	Mach. Screw	I	0.250	0.241	0.076		0.011	5	80
3*	(5) 0.125-40UNC	Mach. Screw	I	0.250	0.241	0.080		0.011	5	120
4*	(6) 0.138-32UNC	Mach. Screw	I	0.250	0.241	0.080		0.012	7	130
5	(6) 0.138-32UNC	Mach. Screw	II	0.312	0.302	0.092		0.013	9	130
6	(8) 0.164-32UNC	Mach. Screw	II	0.344	0.332	0.099		0.013	11	150
7	(10) 0.190-32UNF	Mach. Screw	II	0.375	0.362	0.105		0.013	15	150
8	(10) 0.190-32UNC	Mach. Screw	II	0.375	0.362	0.105		0.017	18	210
9*	(12) 0.216-24UNC	Mach. Screw	III	0.438	0.423	0.118		0.018	27	230
10*	(14) 0.242-24UNC	Mach. Screw	III	0.500	0.489	0.125	±0.010	0.019	- -	- - -
11*	(1/4) 0.250-20UNC	Heavy	III	0.500	0.489	0.134		0.023	38	400
12	(1/4) 0.250-20UNC	Regular	III	0.438	0.428	0.123		0.021	38	350
13	(1/4) 0.250-28UNF	Regular	III	0.438	0.428	0.118		0.016	30	250
14*	(5/16) 0.3125-18UNC	Heavy	III	0.562	0.547	0.145		0.024	70	470
15	(5/16) 0.3125-18UNC	Regular	III	0.500	0.489	0.134		0.021	52	450
16	(5/16) 0.3125-24UNF	Regular	IV	0.500	0.489	0.129		0.019	45	320
17*	(3/8) 0.375-16UNC	Heavy	III	0.688	0.675	0.154		0.027	85	800
18	(3/8) 0.375-16UNC	Regular	III	0.562	0.551	0.145		0.021	70	500
19	(3/8) 0.375-24UNF	Regular	IV	0.562	0.551	0.140		0.019	65	420
20*	(7/16) 0.4375-14UNC	Heavy	III	0.750	0.728	0.168		0.030	180	900
21*	(7/16) 0.4375-20UNF	Light	IV	0.625	0.600	0.150		0.021	85	700
22*	(1/2) 0.500-13UNC	Heavy	III	0.875	0.850	0.202		0.033	210	1100
23	(1/2) 0.500-13UNC	Regular	III	0.750	0.736	0.179		0.027	145	900
24	(1/2) 0.500-20UNF	Regular	IV	0.750	0.736	0.172		0.023	130	750
25*	(9/16) 0.5625-18UNF	Regular	IV	0.875	0.861	0.193		0.024	180	900
26*	(5/8) 0.625-11UNC	Heavy	III	1.062	1.031	0.212	±0.015	0.036	330	1500
27	(5/8) 0.625-11UNC	Regular	III	1.000	0.969	0.224		0.035	370	1500
28	(5/8) 0.625-18UNF	Regular	IV	0.938	0.922	0.203		0.024	200	1000
29*	(3/4) 0.750-10UNC	Heavy	III	1.250	1.212	0.269		0.040	640	1650
30	(3/4) 0.750-10UNC	Regular	III	1.125	1.088	0.246		0.038	530	1650
31	(3/4) 0.750-16UNF	Light	IV	1.062	1.031	0.225		0.027	350	1150
32*	(7/8) 0.875-9UNC	Heavy	III	1.438	1.394	0.290		0.044	- - -	- - - -
33	(7/8) 0.875-9UNC	Regular	III	1.312	1.269	0.280	±0.020	0.042	760	1700
34	(7/8) 0.875-14UNF	Light	IV	1.250	1.207	0.256		0.030	500	1400
35	1.000-8UNC	Heavy	III	1.625	1.575	0.336		0.049	- - -	- - - -
36*	1.000-14UNF	Light	IV	1.437	1.387	0.280		0.030	750	1600
37	1.000-12UNF	Light	III	1.438	1.387	0.280		0.030	- - -	- - - -

* Inactive for new design after 19 JAN 1976.

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REQUIREMENTS:

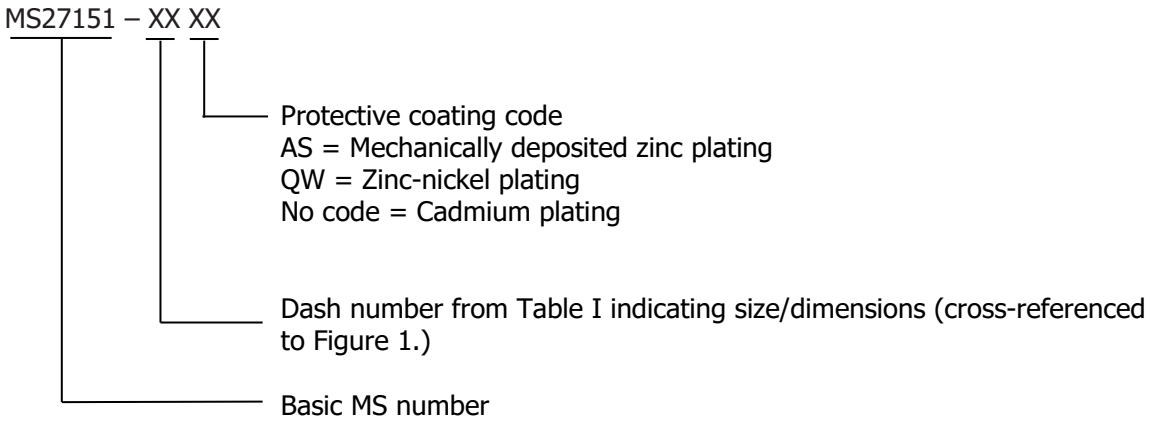
1 Material. Steel, carbon (spring steel), SAE 1050 thru 1066 in accordance with SAE J403. Hardness shall be Rockwell 30N 56-66.

2 Protective finish. Add one of the following codes as specified in the contract, after the dash number to designate protective finish:

AS	-	Zinc, Mechanically Deposited per ASTM B695, Type II, Class 5 with conversion treatment free of hexavalent chromium and meets the Type II corrosion resistance.
QW	-	Zinc-Nickel plate, Type K20 per MIL-PRF-32647.
No code	-	Cadmium plate 1/, per SAE AMS-QQ-P-416 Type II, Class 2.

1/ Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. For ground vehicle applications, both designator QW and AS are suitable alternative finishes.

3 Part or Identifying Number (PIN). The PIN shall consist of the MS number, the dash number, plus the finish code.



PIN examples:

MS27151-1AS is a mechanically deposited zinc finished, carbon steel, stamped nut for use with a 1/4 inch bolt diameter.

MS27151-5QW is a zinc-nickel plated, carbon steel, stamped nut for use with a 1/4 bolt diameter.

MS27151-9 is a cadmium plated, carbon steel, stamped nut for use with a 7/8 bolt diameter.

4 Threads. These nuts are intended for use with threads in accordance with Screw-Thread Standards per ASME B1.1.

5 Workmanship. Stamped nuts shall be free from cracks, burrs, splits, loose scale, or any defects which may affect their serviceability.

6 Dimensions. Unless otherwise specified, all dimensions are in inches.

7 Design. For design feature purposes, this document takes precedence over all other documents.

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8 Documents. Referenced documents shall be of the issue in effect on the date of the invitation for bids unless otherwise specified in the contract. Copies of military specifications are available online at https://quicksearch.dla.mil. Copies of ASME documents are available online at https://www.asme.org. Copies of SAE documents are available online at https://www.sae.org.

9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – YD
Air Force – 99
DLA – GS

Review activities:
Army – AR, AV, CR4, MI
Navy – AS, OS, SH

Preparing Activity:
Army – AT

(Project 5310-2022-016)

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